Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/21/2019
PI Name:	Blaber, Elizabeth Ph.D.		
Project Title:	Osteogenic Differentiation of Somatic Stem Cells in Space: A Study Investigating the Role of CDKN1a/p21 on Mesenchymal Stem Cell Proliferation, Differentiation, and Regeneration in Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6548
Organization Name:	Rensselaer Polytechnic Institute		
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Zip Code:	12180	**Congressional District:**	20
Comments:	NOTE: Moved to Rensselaer Polytechnic Institute in fall 2019 from NASA Ames Research Center.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	07/15/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 7/15/2019 as PI moved to Rensselaer Polytechnic Institute and a new grant 80NSSC20K1125 was issued for continuation at RPI (Ed., 8/27/20) NOTE: End date changed to 12/31/2020 per F. Hernandez/ARC (Ed., 6/20/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Almeida, Eduardo Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight factors, including microgravity and space radiation, are known to affect somatic stem cell-based tissue regeneration and cause significant degenerative conditions in multiple mammalian tissues, posing a significant risk for long-term human space exploration. Our previous spaceflight research has shown an inability of both embryonic and somatic stem cells to proliferate, differentiate, and regenerate damaged cells in microgravity conditions, which may be the cause of the widespread tissue degeneration observed in space. Our laboratory has recently published key evidence that spaceflight induces over expression of CDKN1a/p21, a master cell cycle and tissue regeneration inhibitor, in proliferating osteoprogenitor cells, potentially providing a molecular mechanism for spaceflight tissue regenerative deficits. In this proposal, we seek to test the hypothesis that the key molecular mechanism causing the degenerative effects of microgravity on adult stem cells is activation of cell cycle arrest and impairment of differentiation via CDKN1a/p21 overexpression. We specifically hypothesize that in microgravity, normal mouse bone marrow mesenchymal stem cells with the CDKN1a/p21 gene will be inhibited from proliferating and differentiating normally relative to 1g, but that mesenchymal stem cells from CDKN1a/p21-null mice will show unrestrained proliferation and differentiation that is not sensitive to microgravity conditions. To test this hypothesis, we are proposing a spaceflight experiment to Section B.2 ‘Mammalian Cell, Tissue and Organ Generation and Degeneration in Space’ of the NASA Research Announcement NNH14ZTT001N. We specifically propose to conduct an experiment in the NASA Bioculture System Hardware in which the stromal stem cell population isolated from bone marrow of CDKN1a/p21-null and wildtype mice will be taken to the International Space Station (ISS) National Laboratory frozen on the surface of collagen-coated alginate microcarrier beads, injected into bioreactors, and cultured in osteogenic medium to induce differentiation and mineralized bone nodule formation in a 30 day culture. The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space. This research addresses NASA’s and the National Research Council's (NRC’s) Decadal Survey call to study the molecular and cellular mechanisms of bone loss and to conduct basic research into the mechanisms of bone adaptation to microgravity using transgenic animal models. It also will address the call to understand the precise signaling mechanisms responsible for reduced bone formation in space. This research is highly significant, as it proposes to investigate a molecular mechanism, based on previous spaceflight results, that has the potential to explain and mitigate a broad range of microgravity induced tissue degenerative health conditions. NOTE: Project continues with a new grant 80NSSC20K1125 issued to Rensselaer Polytechnic Institute with the same Principal Investigator Dr. Elizabeth Blaber due to her move to RPI (Ed., 8/27/20)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of this experiment will conclusively determine the role of CDKN1a/p21 in stem cell regenerative health in space. Specifically, if we observe inhibition of osteogenesis by microgravity in wildtype stem cells, but not in CDKN1a/p21-null cells, this will support our hypothesis that the induction of cell cycle arrest by microgravity in adult stem cells is a key molecular mechanism causing the observed degenerative effects of space on mammalian tissues. This will provide a new perspective on the problem of stem cell regenerative health and medicine in space as well as on Earth, and will provide fundamental insight into stem cell regenerative mechanisms that will aid our understanding of basic stem cell functions on Earth and in space.
Task Progress:	Spaceflight factors, including microgravity and space radiation, are known to affect somatic stem cell-based tissue regeneration and cause significant degenerative conditions in multiple mammalian tissues. This poses a significant risk for long-term human space exploration. Our previous spaceflight research has shown an inability of both embryonic and somatic stem cells to proliferate, differentiate, and regenerate damaged cells in microgravity conditions, which may be the cause of the widespread tissue degeneration observed in space. Our laboratory has recently published key evidence that spaceflight induces over expression of CDKN1a/p21, a master cell cycle and tissue regeneration inhibitor, in proliferating osteoprogenitor cells, potentially providing a molecular mechanism for spaceflight tissue regenerative deficits. In this proposal, we hypothesize that CDKN1a/p21 inhibits the proliferation and differentiation of mesenchymal stem cells into bone forming osteoblasts in-vitro in space, and therefore bone marrow mesenchymal stem cells from CDKN1a/p21-null mice will show unrestrained proliferation and differentiation in microgravity. During initial ground-based definition phase of this grant we characterized the ability of bone marrow stem cells isolated from juvenile (11 week old) and skeletally mature (18 week old) female KO mice to differentiate into the osteoblast lineage. We found that cells without the CDKN1a/p21 gene from both juvenile and adult knock out mice exhibited increased bone formation potential as shown by increased proliferation rates and increased mineralized nodule formation. This increase in bone formation potential in-vitro was also seen in-vivo in juvenile mice but not in adult mice. Due to a delay in flight opportunities and hardware restriction, we modified the existing grant to achieve the same science goals but expand our ground-based definition studies to further characterize the role of CDKN1a/p21 on bone formation following increased mechanical loading and during aging. We found that CDKN1a/p21 is a load sensitive molecule due to the increased proliferation and mineralization observed during mechanical loading. Additionally, we found that CDKN1a/p21 plays a significant role in the arrest of bone formation during aging as evidenced by increased proliferation, differentiation, and mineralization potential seen in aged knockout mice. During this time, we have also continued to optimize the experimental parameters that will be used for the spaceflight studies. This included optimization of freeze/thaw conditions, optimization of small molecule slow release beads to maintain consistent levels over the course of the experiment, optimization of feeding schedules and media composition, and optimization of cell/bead numbers. NOTE: Project continues with a new grant 80NSSC20K1125 issued to Rensselaer Polytechnic Institute with the same Principal Investigator Dr. Elizabeth Blaber due to her move to RPI (Ed., 8/27/20)
Bibliography Type:	Description: (Last Updated: 04/03/2025)

Abstracts for Journals and Proceedings	Cheng-Campbell M, Stimpel O, Juran C, Almeida EAC, Blaber EA. "CDKN1a/p21 plays a critical role in suppressing stem cell regenerative potential during aging." Presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Blaber EA, Schreurs A. "Spaceflight-induced Osteoarthritis: A Potential Dietary Countermeasure." Presented at the ASBMR 2018 (American Society for Bone and Mineral Research), Montreal, Canada, September 28-October 1, 2018. J Bone Miner Res 2018;33 (Suppl 1). , Oct-2018
Abstracts for Journals and Proceedings	Cheng-Campbell M, Almeida EAC, Blaber EA. "Role of CDKN1a/p21 in Mouse Bone Turnover and Aging." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Putman E, Cheng-Campbell M, Stimpel O, Almeida EAC, Blaber EA. "The role of CDKN1a/p21 in cellular senescence of bone marrow stem cells under spaceflight stressors." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Stimpel O, Almeida EAC, Blaber EA. "The effects of CDKN1a/p21 on oxidative stress and mitochondrial function during long duration spaceflight." Presented at the 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. 33rd Annual Meeting of the American Society for Gravitational and Space Research, Seattle, WA, October 25-28, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Chen A, Chen J, Cheng-Campbell M, Blaber EA, Almeida EAC. "Effects of Uniaxial Strain on Stem Cells from Wildtype and CDKN1a/p21 Knockout Mice." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Articles in Peer-reviewed Journals	Blaber EA. (Editor-in-chief for the special issue) "Special Issue: Stem Cells, Radiation and Microgravity." Stem Cells and Development. 2018 Sep 15;27(18):1227-9. https://doi.org/10.1089/scd.2018.29003.bla ; PubMed PMID: 30212289 , Sep-2018